

Manish Kumar

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Summary

Machine Learning Engineer with 3+ years of experience designing and deploying end-to-end ML and generative AI systems, including retrieval-augmented generation (RAG), LLM fine-tuning, and scalable MLOps pipelines. Proven ability to productionize NLP solutions with a focus on automation, reliability, and experimentation. Passionate about building agentic, data-driven workflows that drive operational efficiency and support informed decision making.

Experience

Machine Learning Associate

Aug 2025 – Dec 2025

Vector Institute

- Leading development of the **Financial Wellness AI Agent** by integrating RAG pipelines with proprietary enterprise tools, leveraging **Ollama** for LLM orchestration and **Qwen-3** embeddings for high-precision document retrieval.
- Curated financial literacy datasets from **1M+** transaction records, **30k+** annotated chatbot logs, and **250+** diverse publications, enhancing structure and semantic consistency through preprocessing and anonymization.
- Implemented dynamic, **user-centric prompt injection** within the RAG pipeline using custom tools, enhancing context engineering to generate personalized financial guidance.

Classroom Ambassador - Data Science

Sept 2024 – April 2025

Tech2U, University of Toronto

- Developed a scalable multimodal **RAG pipeline** for automated information retrieval from large-scale text and image datasets, integrating a ColPali-based vision-language neural retrieval module.
- Achieved **10× faster retrieval** by optimizing ColPali via **4-bit QLoRA** fine-tuning, custom **32× patch-reducing pooling**, and high recall **re-ranking** achieving **~93% NDCG@10**.
- Engineered a **kernel-based embedding fusion** method extending late chunking, along with a **semantic-aware document parsing** pipeline to improve LLM grounding on long-form context.
- Collaborated with subject-matter experts to design domain-specific chatbot behavior using **instruction-tuned prompt engineering** with **OpenAI APIs**; deployed **FastAPI endpoints** for seamless front-end chat UI integration.

Research Assistant

Oct 2019 – Oct 2022

Artificial Intelligence and Robotics Lab

- Researched **surrogate ML models** to estimate individual control surface moment contributions using multivariate nonlinear regression, **achieving <5% relative error** against wind tunnel data across diverse flight envelopes.
- Developed **physics-informed ML models** integrating 6DOF dynamics to simulate aircraft behavior, enabling real-time closed-loop testing with **sub-millisecond latency**.
- Designed and deployed **CI/CD-driven simulation pipelines** for Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) testing, **reducing system validation and prototyping cycles by 40%**.
- Engineered **ELT pipelines** to clean high-volume, noisy wind tunnel data and built **visualization dashboards** that revealed actionable insights, leading to 8% improvement in aircraft roll performance.

Education

University of Toronto – Master of Engineering | GPA: 3.84/4

Jan 2023 – Jan 2025

Relevant Coursework: Machine Learning, Neural Networks & Deep Learning, Reinforcement Learning, Computer Vision, Cloud-Based Data Analytics, Prompt Engineering

Skills

- **Programming Languages:** Python, SQL
- **Tools & Libraries:** PyTorch, Hugging Face Transformers, Langchain, LangGraph, FastAPI, Git, Docker, MLflow, Scikit-Learn, Streamlit, Model Context Protocol (MCP), RAGAS
- **Technical Skills:** LLM fine-tuning, RAG pipelines, Generative AI, LLM evaluation, Distributed training, Agentic AI
- **Cloud Platforms:** Azure (ML, Databricks, Cognitive Search), AWS SageMaker, GCP (Cloud Run, VMs)
- **Databases:** ChromaDB, Qdrant, FAISS, MongoDB, MySQL, PostgreSQL

Projects

Transformer-Based Text Summarization Microservice ([Portfolio](#))

- **Fine-tuned Google Pegasus** model on Samsum dataset using **Hugging Face Trainer**, achieving a 12% ROUGE-L improvement and enhancing domain-specific short-text abstraction.
- Developed a full-stack MLOps pipeline covering data ingestion, model training, evaluation, and CI/CD integration using **Docker** and **GitHub Actions**, reducing deployment time by 50%.
- Deployed the model as **RESTful microservice on Azure VMs** with **FastAPI** backend and **Streamlit** UI, enabling real-time summarization and feedback-driven model iteration.

LLM-Powered Financial Sentiment Analysis & Trading System ([Portfolio](#))

- **Fine-tuned BERT-based model** on the Financial PhraseBank dataset for sentiment analysis and benchmarked its performance against **FinBERT** to evaluate classification accuracy on domain-specific financial text.
- Built **ELT pipelines** to curate 3,300+ days of financial data; conducted time series analysis and engineered sentiment-aware features to support predictive modeling.
- Designed an **AutoML** workflow integrating Ridge, Random Forest, and Gradient Boosting models; achieved a 40% improvement over baseline via automated model selection and hyperparameter tuning.

Cloud-Based Intrusion Detection System (IDS)

- Implemented ChiSqSelector for feature selection and trained multiple ML models using **distributed training** in **Apache Spark**, achieving AUROC: 99.55%, AUPR: 96.24% on KDD99 dataset.
- Built a reproducible, end-to-end **MLOps workflow** with **Azure Databricks**, **MLflow**, and **Docker**, enabling automated experiment tracking and seamless model portability across environments.

Lightweight CNN for Image & Video Deblurring ([Portfolio](#))

- Designed a compact CNN-based Nested-ResNet (<9M params) using **PyTorch** based on extensive literature review; developed **layer-wise adversarial training**, outperforming DeblurGAN-V2 (60M) by 1.2 dB.
- Scaled training pipeline using **distributed training** (with mixed precision) across 2 heterogeneous GPUs, **enabling 200% larger batch size** and boosting overall training efficiency.
- Experimenting (on-going) blur-conditioned **Latent Diffusion** (multi-scale guidance) and **self-supervised perceptual similarity metrics** to improve out-of-domain generalization.